Aayush Ranjan

cs25mtech11002@iith.ac.in | aayushranjan.dev@gmail.com | +91 8090542099 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.04
Maharishi Vidya Mandir	Class XII (CBSE)	2019	86.40%
Maharishi Vidya mandir	Class X (CBSE)	2017	10.00%

SKILLS

- **Languages:** Python, C/C++, SQL
- **ML / Deep Learning:** PyTorch, Scikit-learn, XGBoost, OpenCV, CNNs
- **NLP / LLMs:** LLM Fine-tuning, LLM Evaluation & Benchmarking, LLM Alignment, LangChain, LangGraph, RAG
- **Libraries & Tools:** NumPy, Pandas, Matplotlib, Git, Streamlit
- **CS Fundamentals:** Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS

PROFESSIONAL EXPERIENCE

EXL Services -- Data Scientist | Banking & Finance Analytics

- Built frequency and severity ML models on historical insurance claims data to estimate **RBNS and IBNR reserves**
- Integrated claim settlement-delay prediction into the reserving pipeline to improve reserve estimation accuracy
- Adapted the pipeline to real-world multi-transaction claims datasets, improving robustness for deployment in reserve-estimation workflows

PROJECTS

Image Denoising & 2× Super-Resolution | KLA AI Hackathon *PyTorch, HAT Transformer, OpenCV*

- Built a joint denoising and **2× super-resolution pipeline,** adapting ImageNet-pretrained RGB weights to single-channel grayscale inputs
- Modelled the degradation as box downsampling with multiplicative Gamma speckle noise and generated synthetic low-resolution samples on the fly for more varied training data
- Trained with a composite Charbonnier + MS-SSIM + FFT-L1 loss, EMA weights and bfloat16 mixed precision; used multi-scale overlap-tile inference with 8-fold test-time augmentation, reaching a final score of **0.88076**

Binary Classification of Synthetic 3D Scenes | IITH DL Hackathon *PyTorch, OpenCV, scikit-learn*

- Identified that labels depended on shape co-occurrence (cube + sphere), explaining why RGB-only models overfit to colour/texture shortcuts
- Built a **dual-stream ResNet-34 trained from scratch** on RGB images and Canny + Sobel gradient features, with MixUp and Stochastic Weight Averaging
- Applied pseudo-labelling, BatchNorm test-time adaptation, 4-view test-time augmentation and an F1-tuned decision threshold

Agentic AI Chatbot with RAG & Tool Calling *LangGraph, LangChain, Groq (Llama 3.1), Ollama, Streamlit, SQLite*

- Built a **LangGraph agent** that dynamically routes queries between direct LLM answers and tools -- PDF retrieval (RAG), DuckDuckGo web search, Alpha Vantage stock API and a calculator
- Implemented thread-scoped RAG over uploaded PDFs (chunking with overlap, nomic-embed-text embeddings, top-k vector similarity search) with persistent multi-session chat memory via a SQLite checkpointer
- Developed a Streamlit UI with token-level streaming, live tool-status indicators, multi-thread chat management and user feedback logging to SQLite

PUBLICATIONS

Classification of Power Quality Disturbances Using Stockwell Transform -- IEEE SCEECS 2024, Bhopal | DOI: 10.1109/SCEECS61402.2024.10482166 (Feb 2024)

Performance Comparison for Feature Extraction Techniques for Power Quality Disturbances -- IEEE ICMICA 2024, Kurukshetra | DOI: 10.1109/ICMICA61068.2024.10732115 (May 2024)

POSITIONS OF RESPONSIBILITY

Teaching Assistant -- Dept. of CSE, IIT Hyderabad

- Courses: Introduction to Programming (ID1063), Introduction to AI (ID1050), Advanced DSACS6013)
- Evaluated programming and algorithmic assignments; conducted doubt-clearing and debugging sessions

Placement Coordinator -- Office of Career Services (OCS), IIT Hyderabad

- Coordinating campus placement activities between students, recruiters and the Office of Career Services

SCHOLASTIC ACHIEVEMENTS

Authored **2 research papers in IEEE conferences** on ML-based power quality disturbance classification

Achieved a **LeetCode contest rating of 1795 (Top 8.52% globally)**

Secured **10th rank** in the IIT Hyderabad Deep Learning Hackathon 2026

Secured **13th rank** in the KLA AI Hack x IIT Hyderabad 2026 (Image denoising & super-resolution)

RELEVANT COURSES

Advanced DSA, Linear Optimization, Advance Computer Networks, Network Security, Foundations of Machine Learning, Deep Learning, Visual Computing, Representation Learning, Fraud Analytics

EXTRACURRICULAR

Volunteer Teacher, Light de Literacy NGO --Contributed over 4 years to teaching initiatives at a registered NGO, bridging the education gap for underprivileged children in slum communities lacking access to formal schooling.